

11.7 Translating Real-World Written Descriptions Into Algebraic Inequalities

Lesson Introduction

 Benchmark	MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line.		 Learning Targets	- I can translate a written description into an algebraic inequality. - I can represent an inequality on a number line when given a written description of a real-world context.
 Benchmark Coherence	Building On N/A		Working Toward MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.	
 Essential Questions	- How do you write an inequality from a verbal description? - How do you represent inequalities on a number line? - Why is the real-world context important when determining possible solutions to inequalities?			
 Lesson Narrative	Students translate verbal descriptions representing a real-world situation into an inequality that represents the situation. They also represent inequalities on number lines. Students use the written description to determine which symbol to use and how it should be represented on a number line.			
 Component Summary	11.7.1 Warm-Up (~5 min.)	Estimate how many TVs it will take to cover a jumbotron (SE p. 273)	11.7.4 Try It (5-10 min.)	Practice translating verbal descriptions of real-world situations into inequalities and represent them on a number line (SE p. 276)
	11.7.2 Exploration (10-15 min.)	Explore how to represent inequalities on a number line (SE p. 273)	11.7.5 Your Turn! (5 min.)	Represent two inequalities on number lines and determine possible values (SE p. 277)
	11.7.3 Guided Practice (10 min.)	Translate verbal descriptions of real-world situations into inequalities and represent them on a number line (SE p. 275)	11.7.6 Wrap-Up (~5 min.)	Demonstrate understanding of writing an inequality, determining possible values that satisfy the inequality, and representing the inequality on a number line (SE p. 278)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> <i>Inequality</i> <i>Strict inequality</i>		N/A	(Optional) Locate pictures and videos of the real-world contexts presented in each section. (Optional) Gather information about students' interests to substitute in for some of the real-world contexts.

 Teacher Tips	Common Misconceptions - Students may struggle to interpret inequalities when the variable is on the right. - Students may struggle to determine the minimum and maximum value of the inequality. - Students may struggle translating verbal descriptions when they involve greater than or equal to and less than or equal to, because these are new in this lesson.	Things to Consider - Consider modeling how to read the $\leq$ and $\geq$ symbols at the beginning of the lesson for students. This will be particularly helpful for ELLs. - Discuss “at least” and “at most” to make sure all students understand what they mean in various contexts. - Consider showing pictures or videos to help students understand the context of the problems. For example, in 11.7.3 Guided Practice, show an illustration of the lines on a highway. - If time is an issue, consider assigning 11.7.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 11.7.2 Exploration, have students first complete the questions independently and then discuss their work with a partner. These discussions will promote student confidence and understanding.	MA.K12.MTR.7.1: Real-World Contexts Students make connections between inequalities and real-world scenarios in 11.7.4 Try It and 11.7.5 Your Turn!. Students represent real-world contexts using inequalities and determine reasonable solutions.
 Differentiate Instruction	This lesson provides an opportunity to engage students by editing questions to appeal to students’ interests. Consider using beginning-of-year surveys or informal observation to learn about students’ interests. For example, in question 1 in 11.7.4 Try It, change the item from a bike to an item most of the class wants. Consider providing pictures or videos of the context in 11.7.4 Try It to provide a visual entry point.	
Lesson Notes:		

11.7.1 Warm-Up: Jumbotron Estimation

Component Overview: Students estimate how many TV screens are needed to cover the jumbotron at Doak Campbell Stadium in Tallahassee.



11.7.1 Warm-Up: Jumbotron Estimation

In the U.S., the average LCD TV screen has a width of 3.63 feet (ft) and a height of 2.04 ft.

Doak Campbell stadium has a jumbotron screen with a width of 119.7 ft and a height of 56.7 ft.

1. Estimate how many TV screens can be laid on top of the jumbotron at Doak Campbell stadium.

Answers will vary. $\frac{(119.7)(56.7)}{(3.63)(2.04)} \sim 916$

$119.7 \div 3.63 \sim 32$ or 33 or $56.7 \div 2.04 \sim 27$ or 28 .

$(32 \cdot 27) = 864$, or $(33 \cdot 28 = 924)$

You can fit 864 tv's to 924 tv's.

2. Explain how you found your estimate.

I found the area of the jumbotron and divided it by the area of the LCD TV.



Consider sharing pictures or videos of a jumbotron and a typical LCD TV. This will provide a visual entry point and help students assess the reasonableness of their estimation.



Doak Campbell Stadium is the home field stadium for the college football team at Florida State University. Show an image or video for any student fans in your classroom.

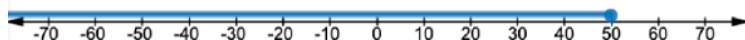
11.7.2 Exploration: Representing Inequalities on a Number Line

Component Overview: Students explore the similarities and differences of solutions to inequalities and strict inequalities. Students analyze a visual representation of how to represent inequalities and strict inequalities on a number line, then practice graphing an inequality and strict inequality on a number line. Students then connect inequalities to real-world scenarios in which they reason about potential solutions and whether they make sense given the real-world context presented.



11.7.2 Exploration: Representing Inequalities on a Number Line

- Complete the following problems.
 - Choose an x -value that will make the inequality $x \leq 50$ true. *Answers will vary. $x = 25$ or $x = 37$*
 - Choose an x -value that will make the inequality $x \leq 50$ false. *Answers will vary. $x = 58$ or $x = 103$*
 - Have your partner verify your choices are correct.
 - What is the x -value that is true for $x \leq 50$ but false for $x < 50$? **50**
 - The number line below represents all of the x -values that are true for $x \leq 50$.

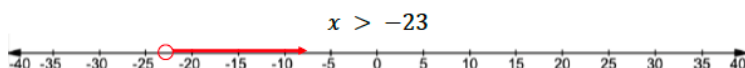
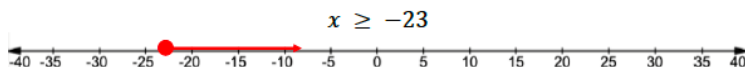


The number line below represents all of the x -values that are true for $x < 50$.



How is the graph for $x < 50$ different from the graph for $x \leq 50$? **$x < 50$ does not include 50.**

- Complete the following.
 - How are the solutions to $x \geq -23$ and $x > -23$ the same?
All values greater than -23 will be solutions.
 - How are the solutions to $x \geq -23$ and $x > -23$ different?
 $x > 23$ does not include -23 .
 - Graph the solutions to each inequality on the number lines.



Think-Pair-Share

Have students first complete the questions independently and then discuss their work with a partner. These discussions will promote student confidence and understanding.



Please read the following inequality: $x \geq 23$

- What does the line under the inequality symbol mean?
- How do you represent this inequality on the number line?

Please read the following inequality: $x > 23$

- How do you represent this inequality on the number line?
- What is the difference between these two inequalities?
- How does the difference impact the solutions to these inequalities?

11.7.2 Exploration: Representing Inequalities on a Number Line (continued)

3. Complete the following.

- a. With your partner, determine which inequality best represents each context.

Nutritional Information <table> <tr> <td>Calories:</td> <td>2,000</td> </tr> <tr> <td>Total Fat</td> <td>Less than 65g</td> </tr> <tr> <td>Sat Fat</td> <td>Less than 20g</td> </tr> <tr> <td>Cholesterol</td> <td>Less than 300mg</td> </tr> <tr> <td>Sodium</td> <td>Less than 2,400mg</td> </tr> </table> Total Fat $f \leq 65$ $f \geq 65$ $f < 65$ $f > 65$	Calories:	2,000	Total Fat	Less than 65g	Sat Fat	Less than 20g	Cholesterol	Less than 300mg	Sodium	Less than 2,400mg	YOU MUST BE AT LEAST 54" TALL TO RIDE $h \leq 54$ $h \geq 54$ $h < 54$ $h > 54$
Calories:	2,000										
Total Fat	Less than 65g										
Sat Fat	Less than 20g										
Cholesterol	Less than 300mg										
Sodium	Less than 2,400mg										
NO MASS GATHERINGS OF MORE THAN 10 PEOPLE $p \leq 10$ $p \geq 10$ $p < 10$ $p > 10$	SPEED LIMIT 50 $s \leq 50$ $s \geq 50$ $s < 50$ $s > 50$										
CAUTION STAY BACK 50 FEET NOT RESPONSIBLE FOR DAMAGE TO VEHICLES OR PERSONS $x \leq 50$ $x \geq 50$ $x < 50$ $x > 50$	APPROX WAIT TIME 35 MIN FROM THIS POINT $w \leq 35$ $w \geq 35$ $w < 35$ $w > 35$ <i>Answers will vary</i>										



Consider opportunities for engaging students with real-world examples of this activity. Does your school or classroom have a collection of signs like these around? Conduct a scavenger hunt. Could students bring in their own nutritional food labels? Conduct a student-led activity on what they bring to class instead.

11.7.2 Exploration: Representing Inequalities on a Number Line (continued)

- b. In which, if any, of these situations does a value of 0 make sense given the context?

Total fat

- c. In which, if any, of these situations does a negative number make sense given the context?

None

- d. In which, if any, of these situations does 1,000 make sense given the context?

None



Have students write similar questions like questions 3a-3d for the other signs they may have collected earlier in 11.7.2 Exploration's discussions. Then have students trade their questions with a partner.

11.7.3 Guided Practice: Translating a Written Description Into an Algebraic Inequality

Component Overview: Students represent real-world scenarios using inequalities and number lines. Then students determine possible solutions to the inequalities. Remind students to consider the context when determining possible solutions.



11.7.3 Guided Practice: Translating a Written Description into an Algebraic Inequality

1. According to the Interstate Highway Standards, U.S. and state highway traffic lanes must be at least 15 feet wide.
 - a. Write an inequality to represent the width, w , that traffic lanes can be. $w \geq 15$

- b. Represent the inequality on the number line.



- c. List two examples of possible widths, in feet, for the traffic lane. *Answers will vary. $w = 15.25$ ft or $w = 15.3$ ft*
 - d. Give one example of a width that is true for the inequality but does not make sense in the real world. *18 ft*

2. High-altitude parachute jumps are usually made for an altitude that is no more than 35,000 feet to avoid frostbite from extreme cold.

- a. If h represents the height of a jumper, write an inequality to describe all possible values of h .
 $h \leq 35,000$

- b. Represent the inequality on the number line.



- c. Does the inequality that represents the situation include 0 as a possible height? *Yes*
 - d. Explain why a negative number does not make sense for this context.
One would not jump with a parachute from a negative height since this is below sea level.



Read each scenario aloud. Then lead a discussion focused on preventing potential misconceptions.



- *What does it mean if something is at least 15 feet wide? Is it larger or smaller than 15? What are three possible solutions?*
- *How do you represent the inequality on a number line?*
- *What does it mean if the jumps are no more than 35,000 feet?*
- *What is the greatest distance possible? What are three possible solutions? How do you represent this on the number line?*

11.7.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

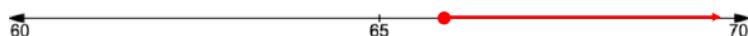
Component Overview: Students independently represent real-world scenarios using inequalities and number lines. Then students determine possible solutions to the inequalities based on the context.



11.7.4 Try It: Writing an Algebraic Inequality and Representing It on a Number Line

- The inequality $67.85 \leq m$ represents the amount of money, m , Edin needs to earn to buy a new bike.

- Graph the inequality.



- Give one value that is true for $67.85 \leq m$ but is unreasonable for the context.
\$700

- Explain why the value you chose is unreasonable.
Edin would have enough money for several bikes.

- A farmer plans to plant more than 52.75 acres of oranges.

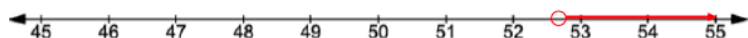
- If A represents the number of acres of oranges the farmer plants, write an inequality that represents the number of acres the farmer plants.
 $A > 52.75$

- Circle all the possible values for A , the number of acres the farmer plans to plant, from the list below.

52.7 acres $52\frac{3}{4}$ acres **53 acres**

52 acres **52.8 acres** $52\frac{1}{4}$ acres

- Graph the inequality.



Consider editing questions or allowing students to adjust contexts to appeal to students' interests. For example, students could adjust question 2 to be "You need to have more than \$52.75 to buy _____."



If students struggle to determine unreasonable values, pose scaffolding questions, such as:
What is the minimum amount that Edin needs?
How much greater or less than the minimum amount would be a reasonable answer?



Challenge students to also determine unreasonable values for the acres too. Ask students to explain why these values are unreasonable based on the context.

11.7.5 Your Turn!

Component Overview: Students work independently to determine if they would rather ride a go-cart or an electric bike based on the speed of each. Students determine possible speeds and plot the given inequalities on number lines.



11.7.5 Your Turn!

1. Would you rather drive a go-cart whose speed, g , in miles per hour (mph), is represented by the inequality $12.5 \geq g$, or an electric bike whose speed, b , in mph, is represented by $b \geq 7\frac{1}{2}$?

a. I choose the ☐ go-cart ☐ electric bike because _____

Answers will vary.

The ☐ go-cart ☒ electric bike could have a speed of 18 mph.

- b. Circle all the values from the list below that are possible speeds for the go-cart.

1.25 mph

15.2 mph

$10\frac{1}{3}$ mph

$\frac{12}{5}$ mph

12.45 mph

11.95 mph

$12\frac{2}{3}$ mph

$11\frac{11}{12}$ mph

- c. Represent the speed of the electric bike on the number line.



- d. Represent the speed of the go-cart on the number line.



- e. Explain why the number lines do not include negative numbers.

You cannot go negative speeds.



Students may struggle to represent the speed of the go-cart on the number line because they have not represented a compound inequality yet. Discuss what negative numbers mean in this context and how they could be represented on the number line.

11.7.6 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate understanding of writing an inequality, determining possible values that satisfy the inequality, and representing the inequality on a number line. Use data from this formative assessment to determine which students need remediation before the next lesson.



11.7.6 Wrap-Up: Ticket Out the Door

At Universal Studios, a person must be at least 48 inches (in.) tall to ride Revenge of the Mummy™.

- Write an inequality representing the height, h , a person must be to go on the ride.

$$h \geq 48$$

- Circle all values from the list below that satisfy this inequality.

47.9 in.

100 in.

$\frac{48}{2}$ in.

48.5 in.

$70\frac{3}{4}$ in.

$48\frac{1}{4}$ in.

0 in.

400 in.

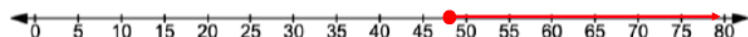
48.099 in.

48 in.

$50\frac{3}{4}$ in.

47.9 in.

- Graph the inequality on the number line.



Consider challenging learners to write an explanation of how they determined their answers (i.e. I tested each value and read the inequality out loud to see if it was true).